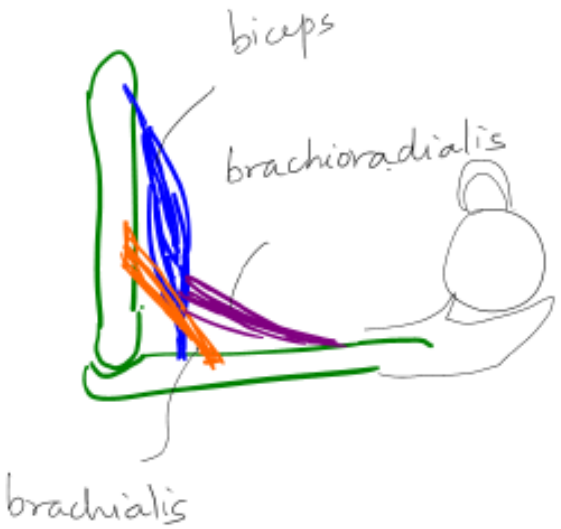
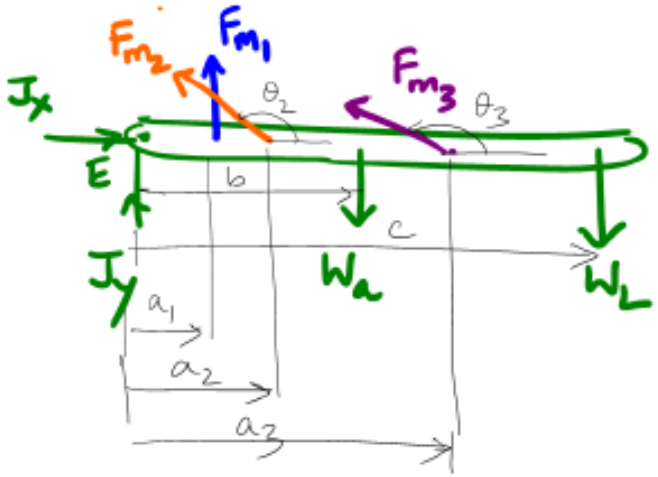


Assume angles of pull for muscles 1, 2, 3 are known, locations of muscle attachments (insertions) are known, lengths of their moment arms are known

$$\sum F_x = 0$$

$$J_x + F_{m_2}\cos\theta_2 + F_{m_3}\cos\theta_3 = 0$$

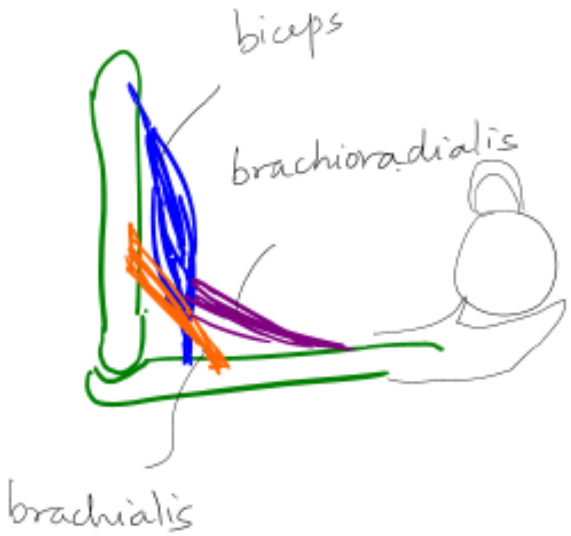


$$\sum F_y = 0$$

$$J_y + F_{m_1} + F_{m_2}\sin\theta_2 + F_{m_3}\sin\theta_3 - W_a - W_L = 0$$

$$\sum M_E = 0$$

$$F_{m_1} a_1 + F_{m_2}\sin\theta_2 \cdot a_2 + F_{m_3}\sin\theta_3 \cdot a_3 - W_a\, b - W_L c = 0$$



Unknowns:

$F_{m_1}, F_{m_2}, F_{m_3}, J_x, J_y$

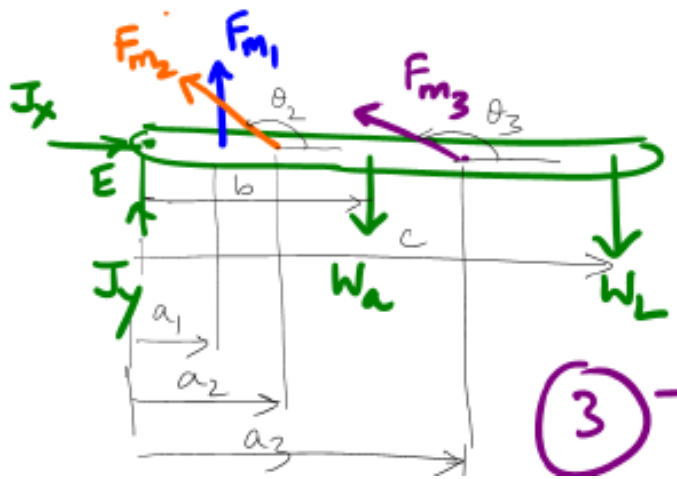
Statically indeterminate system

$$\frac{F_{m_2}}{F_{m_1}} = \frac{A_2}{A_1} = k_{21}$$

$$\frac{F_{m_3}}{F_{m_1}} = \frac{A_3}{A_1} = k_{31}$$

$$\Sigma F_x = 0$$

(1) — $J_x + F_{m_2}\cos\theta_2 + F_{m_3}\cos\theta_3 = 0$



$$\Sigma F_y = 0$$

(2) — $-J_y + F_{m_1} + F_{m_2}\sin\theta_2 + F_{m_3}\sin\theta_3 - W_a - W_L = 0$

$$\Sigma M_E = 0$$

(3) — $F_{m_1} a_1 + F_{m_2}\sin\theta_2 \cdot a_2 + F_{m_3}\sin\theta_3 \cdot a_3 - W_a b - W_L c = 0$

Unknowns reduce to F_{m_1}, J_x, J_y

TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

Strategies for an indeterminate system

(1) Muscle forces proportional to c/s area

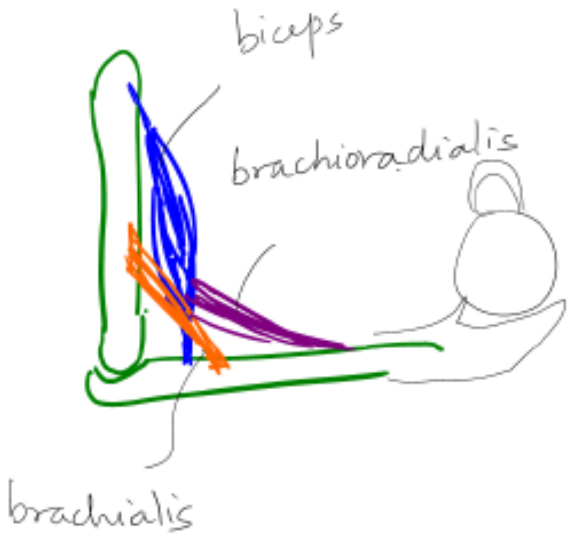
(2) Correlate muscle forces with EMG signal measurements

(3) Optimization algorithm

Minimize muscle forces, joint reactions

Dynamic situation — minimize joint moments, joint powers, etc

(minimum effort)



Unknowns:

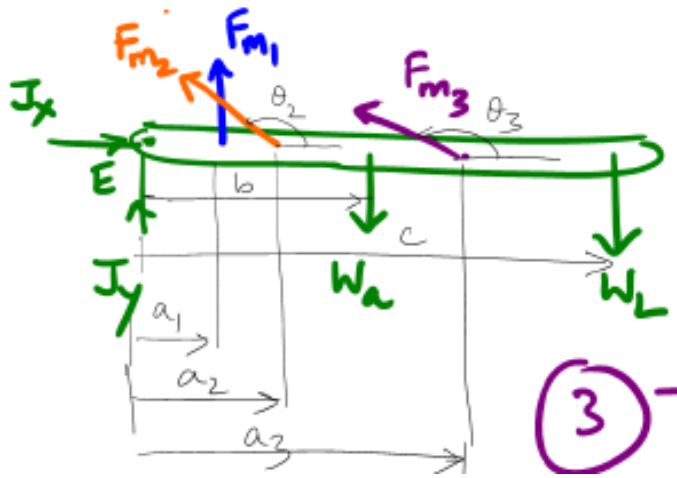
$F_{m_1}, F_{m_2}, F_{m_3}, J_x, J_y$

Statically indeterminate system

$$\frac{F_{m_2}}{F_{m_1}} = \frac{A_2}{A_1} = k_{21} \qquad \frac{F_{m_3}}{F_{m_1}} = \frac{A_3}{A_1} = k_{31}$$

$$\Sigma F_x = 0$$

(1) — $J_x + F_{m_2}\cos\theta_2 + F_{m_3}\cos\theta_3 = 0$



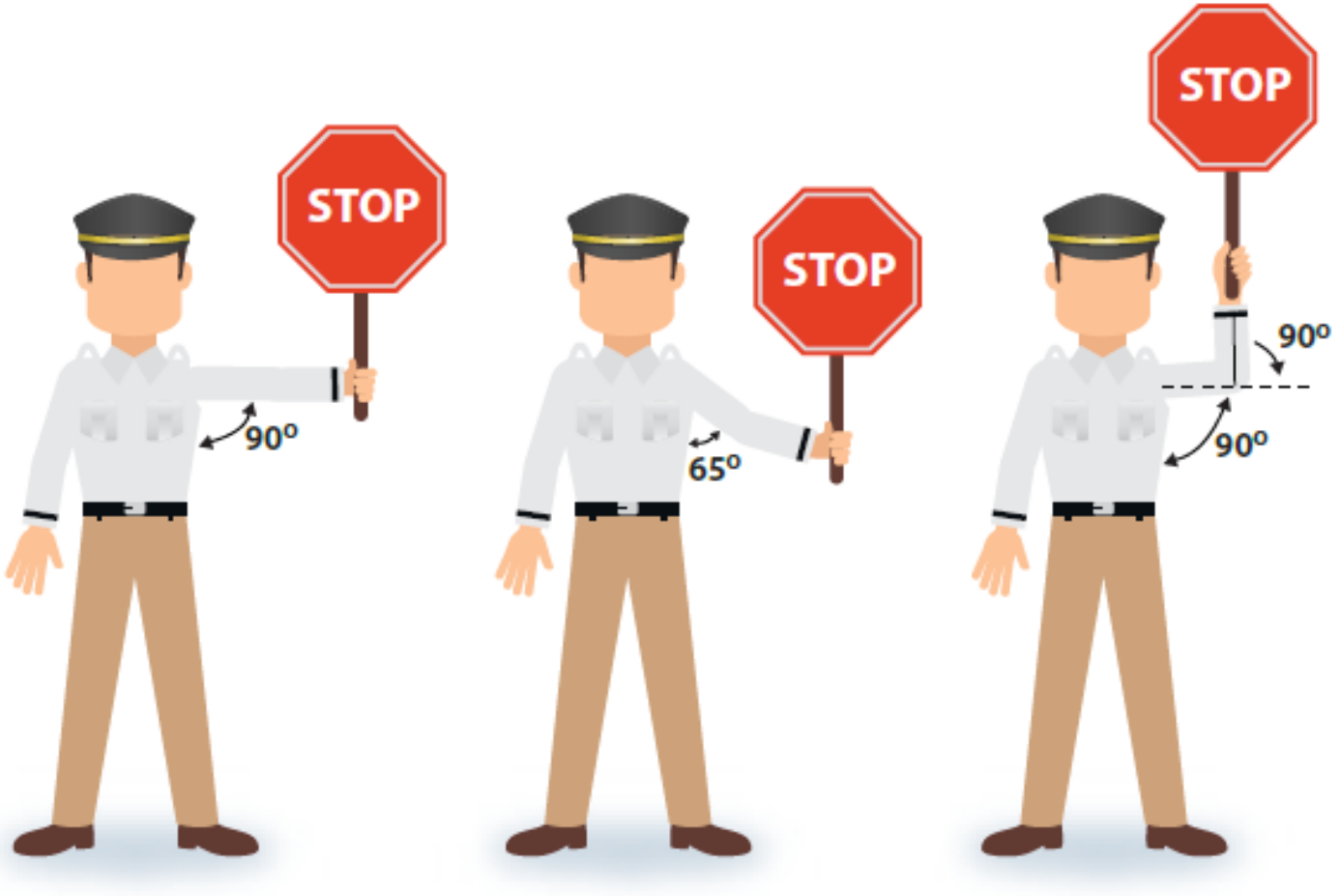
$$\Sigma F_y = 0$$

(2) — $-J_y + F_{m_1} + F_{m_2}\sin\theta_2 + F_{m_3}\sin\theta_3 - W_a - W_L = 0$

$$\Sigma M_E = 0$$

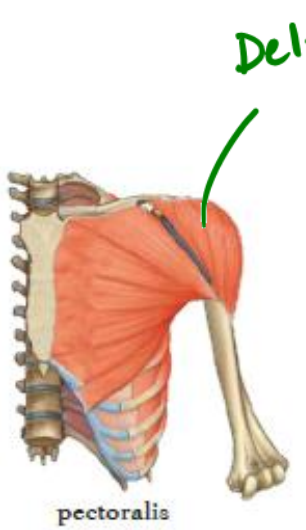
(3) — $F_{m_1} a_1 + F_{m_2}\sin\theta_2 \cdot a_2 + F_{m_3}\sin\theta_3 \cdot a_3 - W_a b - W_L c = 0$

Static analysis - shoulder

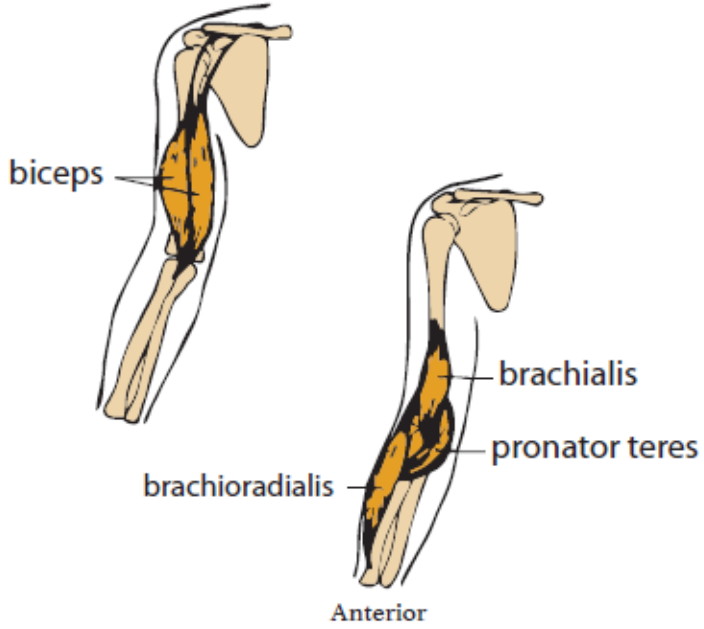


Muscles of the upper limb

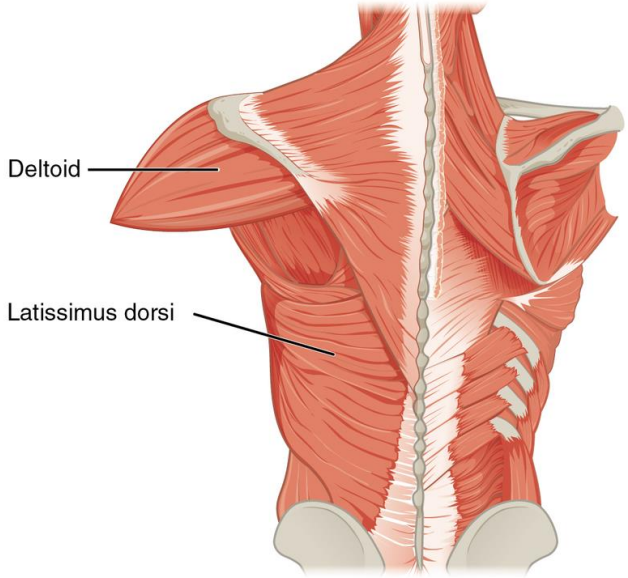






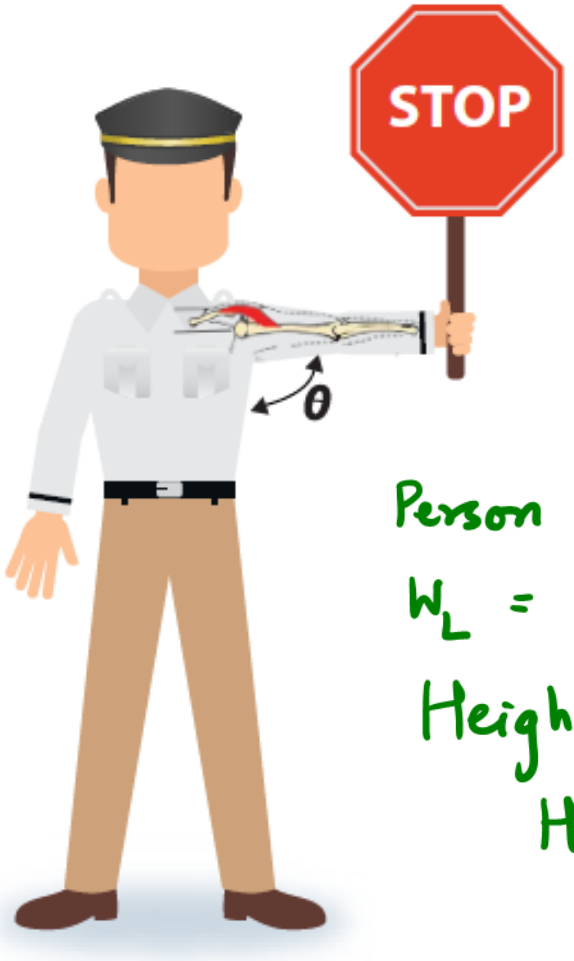


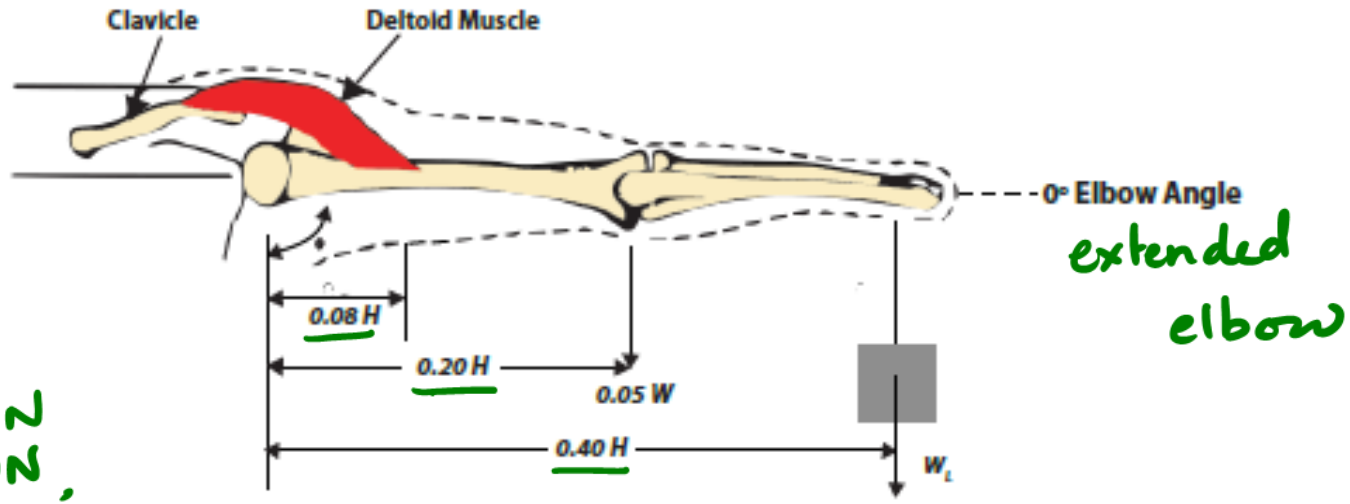
By OpenStax College - https://cnx.org/contents/FPtK1zmh@8.108:y9_gDy74@5, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=64291142



Static analysis - shoulder

Frontal plane





extended elbow

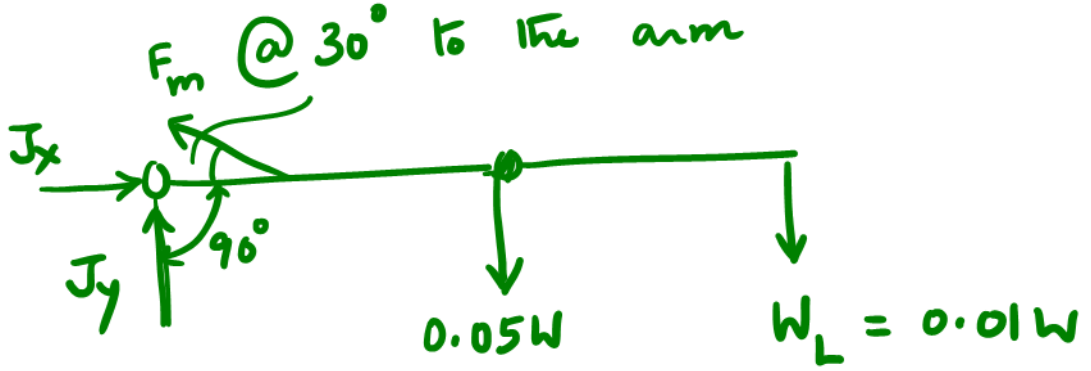
W = 700N

Person of wt = 700N,

$W_L = 0.01\,W$

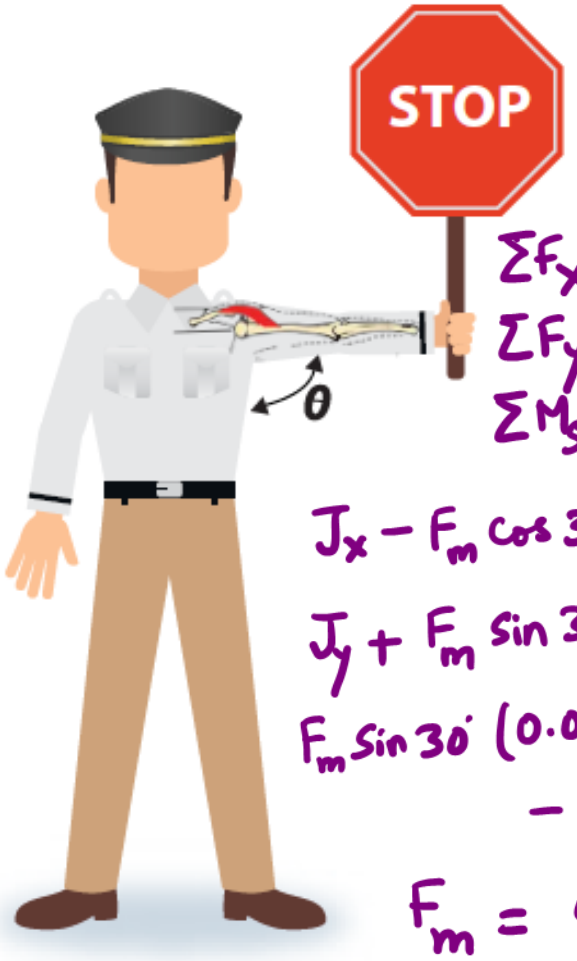
Height of person

H = 1.75m



Static analysis - shoulder

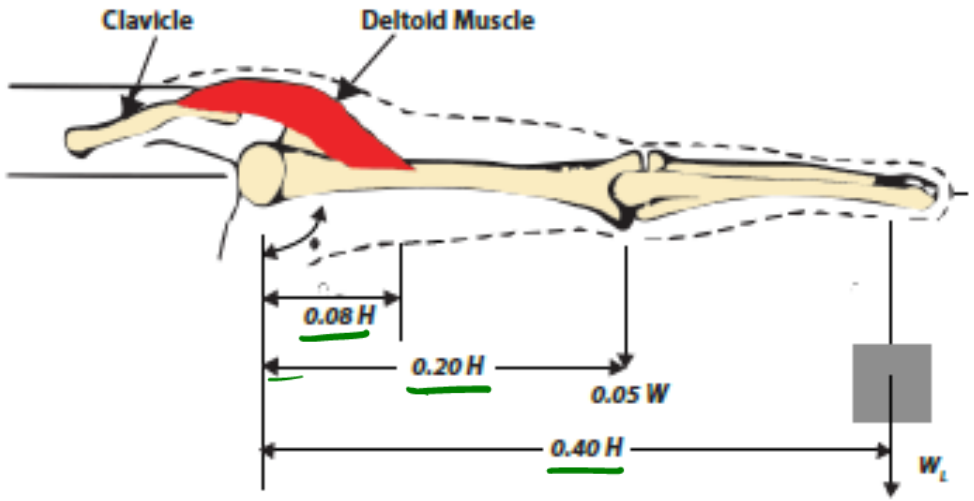
Frontal plane



$\Sigma F_x = 0$

$\Sigma F_y = 0$

$\Sigma M_s = 0$



extended elbow

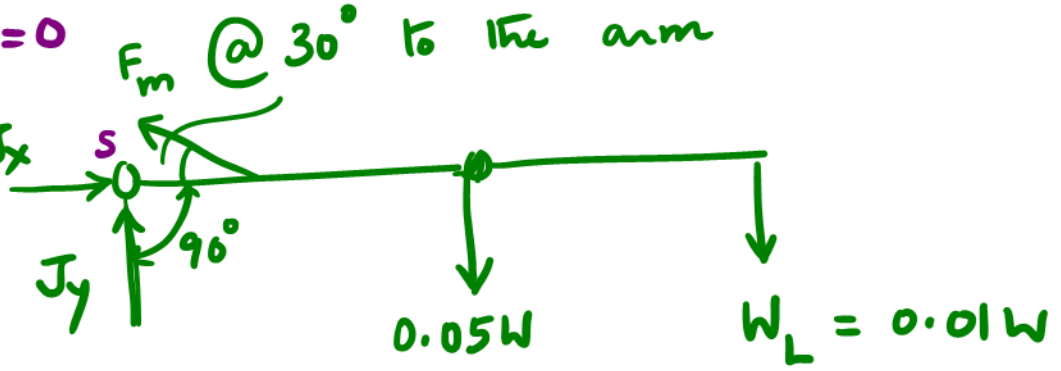
$J_x - F_m \cos 30^\circ = 0$

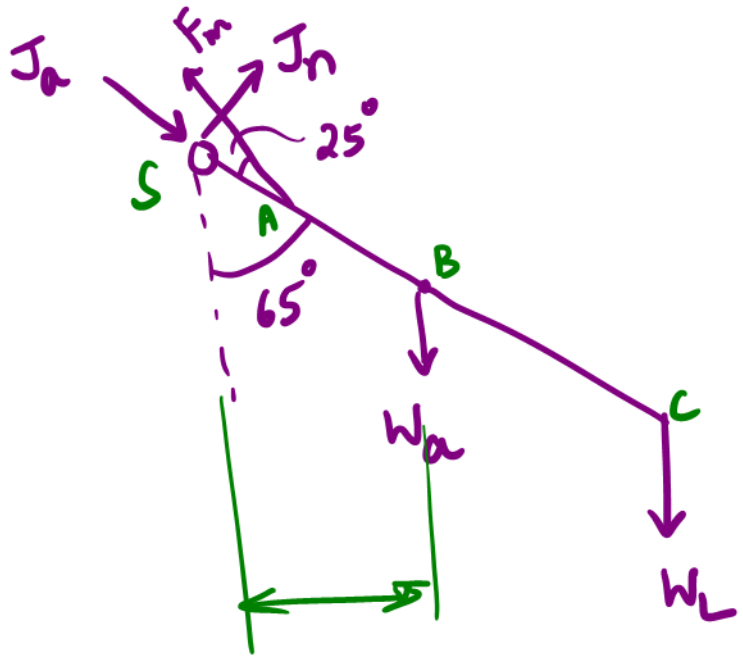
$J_y + F_m \sin 30^\circ - 0.05W - 0.01W = 0$

$F_m \sin 30^\circ (0.08H) - 0.05W(0.2H) - 0.01W(0.4H) = 0$

$F_m = 0.35W = 245\,N$

$J_x = 212.2\,N$, $J_y = -80.5\,N$





Muscle insertion angle is now 25°

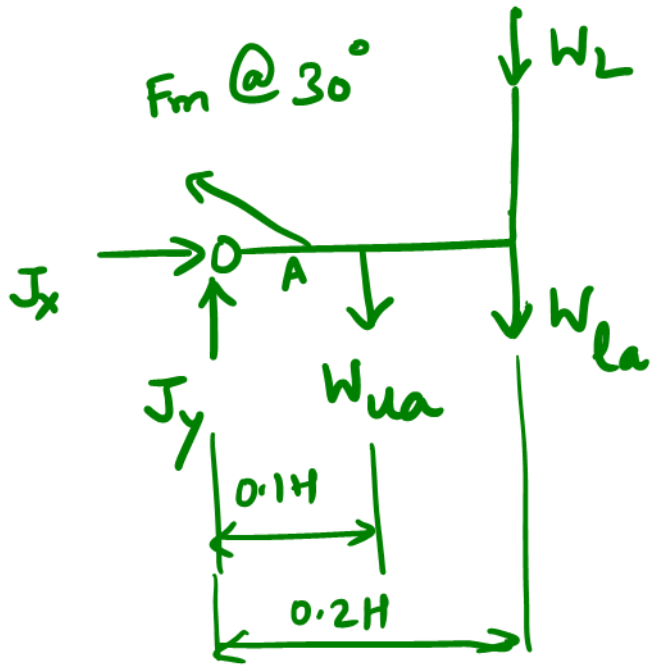
$$\Sigma M_S = 0$$

$$F_m \sin 25^\circ (SA) - W_a (SB \sin 65^\circ) - W_L (SC \sin 65^\circ) = 0$$

$$F_m = \frac{0.05W (0.2H \sin 65^\circ) + 0.01W (0.4H \sin 65^\circ)}{0.08H \sin 25^\circ}$$

$$= 0.375W$$

increased slightly bcos of the disadvantageous angle of insertion



$W_{ua} = 0.03W$ $\quad W_{la} = 0.02W$

$\Sigma M_S = 0$

$F_m \sin 30^\circ (SA) - (W_L + W_{la})\, 0.2H - W_{ua}(0.1H) = 0$

$F_m = 0.225\ W$

36% decrease from Case I where the arm was fully extended at the elbow.

Range of motion @ shoulder

230° flexion - extension

170° abduction - adduction

170° inward - outward rotation